

Hu Zhentao

Multimodal Audio/Video Interaction Engineering / Speech Pipeline Development / Technical AI Product Roles

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TECHNICAL SUMMARY

- Background spans AI product delivery, algorithm research, and engineering integration, with strongest depth in speech interaction pipelines, digital-human driving/rendering integration, AIGC video-generation workflows, tool calling, and time-series signal algorithms.
- Built complete chains from microphone capture, VAD, ASR, LLM/Agent, and TTS to digital-human speaking/render-state synchronization, with practical understanding of low-latency dialogue, interruption handling, turn management, and configurable business logic.
- Current work focuses on user-facing companion interaction prototypes, including speech input/output, synchronization between speech output and expression states, persona design, emotional-consistency control, and real-time interaction experience.
- Able to break user requirements into data, model, API, frontend/backend, deployment, and acceptance checkpoints, and evaluate solution value across latency, cost, delivery complexity, manual maintenance cost, and pilot feasibility.

TECHNICAL STACK

- Audio and interaction: Whisper, VAD, ASR/TTS, Llama3-8B, Qwen, DeepSeek, Tacotron2-DDC-GST, Doubao real-time voice, AEC/3A, Coze Agent.
- Digital humans and video pipelines: OpenChatAvatar, video-generation workflows, anchor-image generation, segmented video generation, programmatic evaluation, resumable execution, subtitle timeline alignment, dubbing synthesis, multilingual export.
- Algorithms and engineering: Python, PyTorch, Transformer, CNN, ResNet, Swin Transformer, ViT, FastAPI, React, Vite, task queues, status polling, result delivery, and cost accounting.

TECHNICAL DEPTH FROM BUSINESS PROJECTS

Early-Stage Startup | Emotional Companion Direction - AI Applications and Audio/Video Technology Projects

2026.01 - Present

1. Wellness Companion Digital Human Project:

- Project summary: For wellness companion scenarios constrained by low willingness to pay and high AI onboarding difficulty, decomposed the product into elderly companionship, memoir generation, scenario Q&A, exception fallback, family linkage, and privacy-authorization modules. Helped secure about RMB 800K in phased budget support and moved the project into commercial-demo and pilot discussions.
- Speech interaction: To address response latency and conversational naturalness in digital-human interaction, explored self-built audio pipelines, open-source pipeline frameworks, and end-to-end interaction models. Connected microphone capture, VAD, ASR, LLM/Agent, TTS, audio playback, and digital-human speaking/render-state synchronization; reduced time to first audio chunk from over **3s** to within **1.5s**.
- Agent development: For companion scenarios where the knowledge base, intents, and tool logic required frequent updates and pure-code maintenance had a high threshold, designed a Coze Agent route. The local audio pipeline handled capture, recognition, synthesis, and playback, while Coze Agent handled companion dialogue, intent recognition, tool routing, knowledge-base maintenance, and business logic. This shifted business iteration from "engineering code changes" to "configuration and workflow adjustments" and reduced intent-recognition time from **2s** to **0.5s**.
- Visual interaction: Built an AIGC video-generation workflow covering script planning, anchor-image generation, segmented video generation, programmatic evaluation, failure retry, and resumable execution, while supporting semi-automated digital-human template generation. Used "video generation + workflow orchestration" to supplement parts of the traditional virtual-engine pipeline, reducing customization barriers and rework cost: per-user custom-avatar cost dropped from **at least RMB 10,000** to **under RMB 1,000**, and delivery time fell from **30 days** to **10 days**.

2. AI Virtual Try-On Project

- For an AI-powered automatic costume-changing requirement around Zhuang ethnic clothing, developed face-swap, person-swap, and FastFit try-on workflows, deployed the service for the project's AIGC module, and resolved engineering issues in Android-side video-stream transport and video compression.

3. AI Comic Project

- For overseas video-translation workflows that were fragmented across manual audio extraction, transcription, translation, subtitle alignment, and dubbing export, designed a **video-translator** pipeline linking audio extraction, ASR transcription, translation polishing, subtitle timeline alignment, dubbing synthesis, and multilingual export, with clear definitions for source video, subtitles, character voice, target language, segmentation strategy, and manual review checkpoints.

4. Cultural Tourism Multi-Agent Project

- For client demos that needed "product introduction + Q&A + recommendation + form capture + tool calling" to run reliably in one chain, participated in orchestration of a multi-agent business advisory assistant, using function calling and prompt engineering to organize Q&A, recommendation, form collection, and tool-invocation flows.

5. Project Management Responsibilities

- To address resource, interface, and vendor-coordination issues as projects moved from prototype to delivery, participated in connecting model platforms, agent tools, and digital-human hardware resources; organized project plans, phased budgets, procurement, vendor, and deployment materials; and handled joint debugging and fixes across service interfaces, runtime state, resource files, and delivery acceptance to reduce delivery risk caused by cross-team information gaps.

- Huawei Consumer BG - Audio Algorithm Engineer Intern**

2024.07 - 2024.10
- For endpoint speech scenarios involving echo cancellation and call-quality validation, studied spectrograms, 3A algorithms, AEC evaluation, and device-side audio-processing methods. Based on ICASSP 2021 [nn-ref](#) work, completed exploratory research, built an algorithm prototype, and collected experimental data.
 - For AEC differences and issue localization across iPhone and Huawei devices, participated in terminal-device AEC testing and accumulated experience in audio algorithm evaluation, device-side comparison, issue localization, and test documentation.

ALGORITHM FOUNDATION AND APPLIED PROJECTS

- Speech Interaction Model Based on Llama3-8B**

2024.10 - 2024.12
- For low-latency speech interaction, built a [whisper_large_v2 + Llama3-8B + tacotron2-DDC-GST](#) ASR-LLM-TTS exploratory chain to validate combined speech recognition, dialogue reasoning, and speech synthesis, with the goal of further reducing interaction latency.
 - For adaptation differences across models in audio-data scenarios, tested Qwen and DeepSeek model families in audio-data fine-tuning settings and used the results to inform later experiment design.

- Human Activity Recognition with mmWave Signals**

2023.06 - 2025.12
- To address the lack of sufficiently large and realistic public datasets for complex daily-action recognition, collected raw mmWave radar data, organized the dataset, defined the experimental plan, and completed thesis experiments. The dataset covered 6 daily activity classes and 20 participants, exceeding the public reference datasets used at the time and providing a more realistic basis for later model comparisons.
 - To evaluate adaptation differences among visual/time-series models on mmWave tasks, reproduced and compared mainstream models and variants including ResNet, Swin Transformer, and ViT, built a multi-dimensional benchmark, and gained hands-on experience in research-oriented experiment design and error attribution.
 - To improve generalization in end-to-end activity recognition, combined traditional signal processing inspired by wavelet transform and fractional Fourier transform with Transformer/CNN deep-learning components, improving model generalization and task adaptability.

EDUCATION

- University of Chinese Academy of Sciences - M.S. in Electronic Information, Institute of Computing Technology**

2022.09 - 2026.01
- Research focus: artificial intelligence, sensor algorithms, time-series signal processing, and LLM applications.
- Xidian University - B.S. in Electronic Information**

2018.09 - 2022.06
- Coursework foundation: image processing, signal processing, signals and systems, random signal analysis, and electromagnetics and wireless-systems courses.

PUBLICATIONS AND OUTCOMES

- Time-domain multi-scale analysis method for wireless human activity recognition, first author; inspired by wavelet transform and using a Transformer-based preprocessing module.
- T-FrFT: Fractional Fourier Transform based Discriminative Feature Extraction Method for Human Activity Classification, IJCNN 2025, first author.